Lecture 2: Practice Problems — Answers and Hints

Linear Algebra

November 14, 2026

Norms

1.1. (a) 9, 5, 4. (b) 7, 5, $337^{1/4} \approx 4.28$, 4: the p-norms decrease towards the ℓ_∞ norm. (c) $\max_i |x_i|^p \le \sum_i |x_i|^p \le n \max_i |x_i|^p$; take p-th roots, and use $n^{1/p} \to 1$.

1.2. (a) A norm; its unit ball is the diamond with vertices $(\pm 1, 0)$ and $(0, \pm\frac{1}{2})$. (b) Not absolutely homogeneous: $f(2\mathbf{e}_2) = 4 \neq 2f(\mathbf{e}_2)$. (c) Not positive definite: $f(1,1) = 0$. (d) A norm (a sum of two functions that satisfy the triangle inequality, and which is 0 only for $\mathbf{x} = \mathbf{0}$). (e) The triangle inequality fails: $f(\mathbf{e}_1 + \mathbf{e}_2) = 4 > 2 = f(\mathbf{e}_1) + f(\mathbf{e}_2)$.

Inner Products

2.1. (a) Yes, $A = \operatorname{diag}(1, 2)$. (b) No: $\langle \mathbf{e}_2, \mathbf{e}_2 \rangle = -1$. (c) Yes, $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ ($2 > 0$, $\det = 1 > 0$). (d) No: $\langle \mathbf{e}_1, \mathbf{e}_1 \rangle = 0$. (e) No: not symmetric. (f) No: $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ for $\mathbf{x} = (1, -2)^\top$, since $\det = 0$.

2.2. (a) $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = 1$, $\|\mathbf{e}_1\| = \sqrt{2}$, $\|\mathbf{e}_2\| = 1$, angle $45°$. (b) $(1, -2)^\top$. (c) $\frac{1}{\sqrt{2}}(1, 0)^\top$ and $\left(-\frac{1}{\sqrt{2}}, \sqrt{2}\right)^\top$.

2.3. (a) $|t| < 2$; $t > 1$. (b) The upper left determinants are 2, 3, 4.

2.5. (a) $L = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 2 \end{pmatrix}$: $L_{11} = \sqrt{4}$, $L_{21} = L_{31} = \frac{2}{2}$, $L_{22} = \sqrt{5-1}$, $L_{32} = \frac{3-1}{2}$, $L_{33} = \sqrt{6-1-1}$. A matrix of the form $LL^\top$ with an invertible L is positive definite, since $\mathbf{x}^\top LL^\top \mathbf{x} = \|L^\top \mathbf{x}\|^2$. (b) $L\mathbf{y} = (8, 10, 11)^\top$ gives $\mathbf{y} = (4, 3, 2)^\top$, and $L^\top \mathbf{x} = \mathbf{y}$ gives $\mathbf{x} = (1, 1, 1)^\top$. $\det(A) = (2 \cdot 2 \cdot 2)^2 = 64$. (c) $\mathbf{x}^\top A \mathbf{x} = \|L^\top \mathbf{x}\|^2 = (2x_1 + x_2 + x_3)^2 + (2x_2 + x_3)^2 + (2x_3)^2$.

2.4. (a) $\operatorname{tr}(SK) = \operatorname{tr}\left((SK)^\top\right) = \operatorname{tr}(K^\top S^\top) = -\operatorname{tr}(KS) = -\operatorname{tr}(SK)$, $\operatorname{sotr}(S^\top K) = \operatorname{tr}(SK) = 0$. (b) Pythagoras for the Frobenius inner product.

Lengths and Distances

3.1. (a) $2x + 3y + 6z \le \|(2, 3, 6)\| \cdot 1 = 7$, attained at $\frac{1}{7}(2, 3, 6)$. (b) Apply Cauchy–Schwarz to $(\sqrt{a}, \sqrt{b}, \sqrt{c})$ and $(\frac{1}{\sqrt{a}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{c}})$; equality for $a = b = c$. (c) $\|(1, 2, 2)\|^2 = 9$.

3.2. (a) $\frac{1}{4}(49 - 25) = 6$. (b) By the parallelogram law, $49 + 25 = 2 \cdot 25 + 2\|y\|^2$, so $\|y\| = 2\sqrt{3}$.

3.3. (a) The difference is $(-3, 2, 6)$: 11, 7, 6, and $\sqrt{9 + 16 + 36} = \sqrt{61}$. (b) Apply the triangle inequality to $x = (x - y) + y$ and to $y = (y - x) + x$.

Angles and Orthogonality

4.1. (a) 90°. (b) $\cos\omega = \frac{4}{9}$, $\omega \approx 63.6°$. (c) $\cos\omega = \frac{1}{\sqrt{3}}$, $\omega \approx 54.7°$. (d) $(1,1,0)$ and $(1,0,1)$: 60°.

4.2. $\operatorname{span}\{(-1,1,1)^\top\}$.

4.3. (a) The columns are orthonormal, and $Q^2 = Q^\top Q = I_3$. (b) $Q\mathbf{x} = \frac{1}{3}(5,1,1)^\top$, with length $\sqrt{3} = \|\mathbf{x}\|$. (c) Q is the reflection in the plane orthogonal to $\mathbf{u}$, i.e., the plane $\mathbf{x} - y - z = 0$; $\det(Q) = -1$.

4.4. (a) 1 and $\frac{1}{\sqrt{60}} \approx 0.13$. (b) $\sqrt{24} \approx 4.90$ and $\sqrt{14} \approx 3.74$. (c) Cosine similarity says d_2, the Euclidean distance says d_3. Since $d_2 = 3d_1$ is the same document repeated three times, the cosine similarity gives the more sensible answer.

Orthonormal Bases

5.1. (a) $\frac{1}{2}(1,1,1,1)$, $\frac{1}{2}(1,1,-1,-1)$, $\frac{1}{\sqrt{2}}(1,-1,0,0)$. (b) $\frac{1}{\sqrt{2}}(0,0,1,-1)$.

5.2. (b) $\langle \mathbf{v}, \mathbf{b}_1 \rangle = 3$, $\langle \mathbf{v}, \mathbf{b}_2 \rangle = 0$, $\langle \mathbf{v}, \mathbf{b}_3 \rangle = 3$, so $\mathbf{v} = 3\mathbf{b}_1 + 3\mathbf{b}_3$; and $3^2 + 0^2 + 3^2 = 18 = \|\mathbf{v}\|^2$.

5.3. Q has the columns $\frac{1}{\sqrt{2}}(1,1,0)^\top$ and $\frac{1}{\sqrt{6}}(1,-1,2)^\top$, and $R = \begin{pmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \frac{\sqrt{6}}{2} \end{pmatrix}$.

Orthogonal Complements

6.1. $U^\perp$ is the null space of the matrix with rows $(1,2,0,1)$ and $(0,1,1,0)$; a basis is $(2,-1,1,0)^\top$, $(-1,0,0,1)^\top$.

6.2. (a) $\operatorname{col}(A)$ is spanned by $(1,2,3)^\top$; $\operatorname{null}(A^\top)$ by $(-2,1,0)^\top$ and $(-3,0,1)^\top$; $\operatorname{row}(A)$ by $(1,2)^\top$; and $\operatorname{null}(A)$ by $(-2,1)^\top$. (b) Exactly when $\mathbf{b}$ is orthogonal to $\operatorname{null}(A^\top)$, i.e., when $\mathbf{b}$ is a multiple of $(1,2,3)^\top$.

6.3. (b) $\mathbf{w}^\top\mathbf{x} + b$ is 2, -1, -1, so the classes are $+1$, -1, -1, and the distances are $\frac{2}{\sqrt{5}}$, $\frac{1}{\sqrt{5}}$, $\frac{1}{\sqrt{5}}$. The point $(1,1)$ has the largest margin.

6.4. $\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1$, i.e., $6x + 3y + 2z = 6$, with normal vector $(6,3,2)$ of length 7; the distance is $\frac{6}{7}$.

Inner Products of Functions

7.1. (a) $\langle 1, x \rangle = \frac{1}{2}$, $\|1\| = 1$, $\|x\| = \frac{1}{\sqrt{3}}$, so $\cos\omega = \frac{\sqrt{3}}{2}$ and $\omega = 30°$. (b) 1 and $x - \frac{1}{2}$. (c) $p(x) = x - \frac{1}{6}$.

7.2. The integrals of $\cos(kx)$ over $[-\pi, \pi]$ vanish for $k \neq 0$; $\sin(mx)\cos(nx)$ is an odd function.

Orthogonal Projections

8.1. $\pi(\mathbf{x}) = (2,2,2)^\top$, $P = \frac{1}{3}\mathbf{1}\mathbf{1}^\top$ (all entries $\frac{1}{3}$), and $\mathbf{x} - \pi(\mathbf{x}) = (0,-1,1)^\top$.

8.2. (a) $\lambda = (\frac{5}{3}, \frac{2}{3})$, $\pi_U(\mathbf{x}) = \frac{1}{3}(7,2,5)^\top$, and the distance is $\frac{4}{\sqrt{3}}$. (b) $\mathbf{n} = (-1,1,1)^\top$, and $\mathbf{x} - \frac{4}{3}\mathbf{n} = \frac{1}{3}(7,2,5)^\top$.

8.3. (a) $C = I_n - P$, where $P = \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ is the projection onto $\mathrm{span}\{\mathbf{1}\}$. (b) $\frac{1}{n}\mathbf{1}\mathbf{1}^\top\mathbf{x} = \bar{x}\mathbf{1}$; $C\mathbf{x} = (-2,-1,0,3)^\top$.

8.4. (a) 1, closest point $\frac{1}{3}(5,-1,1)$. (b) $\frac{|10-1|}{3} = 3$. (c) The projection is $(2,2,2)$, so the distance is $\|(-1,0,1)\| = \sqrt{2}$.

8.5. $y = 0.5 + 0.6x$; residuals $(-0.1, 0.3, -0.3, 0.1)$; sum of squared errors 0.2.

8.6. (a) The normal equation is $Nc = \sum_i y_i$; the fitted vector $\bar{y}\mathbf{1}$ is the projection of $\mathbf{y}$ onto $\mathrm{span}\{\mathbf{1}\}$. (b) The normal equation is $(\sum_i x_i^2)w = \sum_i x_i y_i$.

Rotations

9.1. (a) The columns $\frac{1}{3}(2,2,-1)$, $\frac{1}{3}(-1,2,2)$, $\frac{1}{3}(2,-1,2)$ are orthonormal, and $\det(R) = 1$. (b) $R - I_3$ has the null space spanned by $(1,1,1)^\top$, so $\mathbf{v} = \frac{1}{\sqrt{3}}(1,1,1)^\top$. (c) $\mathrm{tr}(QR_3(\theta)Q^\top) = \mathrm{tr}(R_3(\theta)Q^\top Q) = \mathrm{tr}\, R_3(\theta) = 1 + 2\cos\theta$. Here $\mathrm{tr}(R) = 2$, so $\cos\theta = \frac{1}{2}$ and $\theta = 60°$ (up to the sign, which depends on the orientation of the axis).

9.2. $R_3(\frac{\pi}{2})R_1(\frac{\pi}{2})\mathbf{e}_2 = \mathbf{e}_3$ and $R_1(\frac{\pi}{2})R_3(\frac{\pi}{2})\mathbf{e}_2 = -\mathbf{e}_1$.

9.3. (a) The columns are orthonormal and $\det(C) = 1$. (b) $\mathbf{v} = (1,1,1)^\top$. (c) $\mathrm{tr}(C) = 0 = 1 + 2\cos\theta$, so $\theta = 120°$, and three rotations by 120° give the identity.

9.4. $\cos\theta = \frac{3}{5}$, $\sin\theta = -\frac{4}{5}$, and $r = 5$.

9.5. (b) Multiply out and use the addition formulas for cos and sin.

The Spectral Theorem

10.1. A_1: $\lambda = 4, 2$ with $q = \frac{1}{\sqrt{2}}(1,1)$, $\frac{1}{\sqrt{2}}(1,-1)$; positive definite. A_2: $\lambda = 2, -3$ with $q = \frac{1}{\sqrt{5}}(2,1)$, $\frac{1}{\sqrt{5}}(1,-2)$; not positive definite.

10.2. $A = J + 3I_3$ has the eigenvalues 5, with $\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$, and 2 (twice), with $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1,-1,0)$ and $\mathbf{q}_3 = \frac{1}{\sqrt{6}}(1,1,-2)$.

10.3. (a) $A = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$ has eigenvalues $\pm\frac{1}{2}$, so the maximum is $8 \cdot \frac{1}{2} = 4$ (at $\pm(2,2)$) and the minimum is -4 (at $\pm(2,-2)$). (b) Maximum 3 at $\pm\frac{1}{\sqrt{2}}(1,1)$, minimum -1 at $\pm\frac{1}{\sqrt{2}}(1,-1)$.

10.4. The first matrix $\begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$ has eigenvalues 6 and 1, with eigenvectors $(2,1)$ and $(1,-2)$: an ellipse with semi-axes $\frac{1}{\sqrt{6}}$ along $(2,1)$ and 1 along $(1,-2)$. The second matrix has eigenvalues 3 and -1: in the rotated coordinates, the curve is $3y_1^2 - y_2^2 = 1$, a hyperbola with axes along $(1,1)$ and $(1,-1)$.

10.5. (b) $P = \sum_{\lambda_i=1} \mathbf{q}_i\mathbf{q}_i^\top$, and both $\mathrm{tr}(P)$ and $\mathrm{rank}(P)$ equal the number of eigenvalues 1. (c) $\mathrm{tr}(P) = 1 = \mathrm{rank}(P)$; P projects onto the line $y = x$.

10.6. (a) $D = \mathrm{diag}(1,2,2,1)$, and $L = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$. (b) $(1-2)^2+(2-4)^2+(4-7)^2 = 14$. (c) $Lq = (2-\sqrt{2})q$, so q is the Fiedler vector; its signs split the path into $\{1,2\}$ and $\{3,4\}$.

10.7. (a) Eigenvectors $\frac{1}{2}(1,1,1,1)$ for 0; $\frac{1}{\sqrt{2}}(1,0,-1,0)$ and $\frac{1}{\sqrt{2}}(0,1,0,-1)$ for 2; $\frac{1}{2}(1,-1,1,-1)$ for 4. (b) The ratio cuts are $2\cdot(\frac{1}{2}+\frac{1}{2}) = 2$, $4\cdot 1 = 4$ and $2\cdot(1+\frac{1}{3}) = \frac{8}{3}$, all at least $\lambda_2 = 2$. Equality holds if and only if the vector f of the lecture's theorem on the ratio cut is an eigenvector for λ_2; here $f = (1,1,-1,-1)$ for the first split.

10.8. (a) $d_i = n-1$ and $W = \mathbf{1}\mathbf{1}^\top - I_n$. Eigenvalues: 0 (eigenvector $\mathbf{1}$) and n, with multiplicity $n-1$ (every x with $x^\top\mathbf{1} = 0$). (b) $\mathrm{cut}(A) = |A||\bar{A}|$, so $\mathrm{RatioCut}(A) = |A||\bar{A}|\frac{|A|+|\bar{A}|}{|A||A|} = n = \lambda_2$: all splits are equally good, and the graph has no cluster structure.

10.9. (a) Σ has the eigenvalues 9 and 1, with unit eigenvectors $\frac{1}{\sqrt{2}}(1,1)^\top$ and $\frac{1}{\sqrt{2}}(1,-1)^\top$, so $\Sigma^{1/2} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $\Sigma^{-1/2} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$; indeed $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. (b) The covariance matrix of $\mathbf{z}_i = M\mathbf{x}_i$ is $M\Sigma M^\top$, and with $M = \Sigma^{-1/2}$ (symmetric), $\Sigma^{-1/2}\Sigma^{1/2}\Sigma^{1/2}\Sigma^{-1/2} = I_2$. (c) For example $-\Sigma^{1/2}$, or $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ (eigenvalues 3 and -1). Neither is positive semidefinite; the square root is only unique among positive semidefinite matrices.

10.10. (a) Write $\mathbf{x} = \sum_i y_i\mathbf{q}_i$. The plane U and the plane $\mathrm{span}\{\mathbf{q}_2,\mathbf{q}_3\}$ meet in a line, and a unit vector on it has $\mathbf{x}^\top A\mathbf{x} = \lambda_2 y_2^2 + \lambda_3 y_3^2 \le \lambda_2$. Similarly, U meets $\mathrm{span}\{\mathbf{q}_1,\mathbf{q}_2\}$, where $\mathbf{x}^\top A\mathbf{x} \ge \lambda_2$. (b) For a unit vector, $|\mathbf{x}^\top E\mathbf{x}| = |\sum_{i,j} E_{ij}x_ix_j| \le 0.01(\sum_i |x_i|)^2 \le 0.03$, since $(\sum_i |x_i|)^2 \le 3\sum_i x_i^2$ by Cauchy–Schwarz. So the eigenvalues of E lie in $[-0.03, 0.03]$, and by Weyl's inequality, the eigenvalues of B lie in $[2.97, 3.03]$, $[1.97, 2.03]$ and $[0.97, 1.03]$.

10.11. The components are $\{1,2,3\}$ and $\{4,5\}$, so $\mathrm{null}(L)$ has the basis $(1,1,1,0,0)$, $(0,0,0,1,1)$, and 0 has multiplicity 2. The block of $\{4,5\}$ is $\begin{pmatrix} 3 & -3 \\ -3 & 3 \end{pmatrix}$, with the eigenvalue 6; the block of $\{1,2,3\}$ is $\begin{pmatrix} 2 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 1 \end{pmatrix}$, with the characteristic polynomial $-\lambda(\lambda^2 - 6\lambda + 6)$, so the remaining eigenvalues are $3 \pm \sqrt{3}$.

The Singular Value Decomposition

11.1. $\sigma_1 = 3$, $\sigma_2 = 2$; for example, $U = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\Sigma = \mathrm{diag}(3,2)$, $V = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

11.2. (a) $\sigma_1 = 2$ (and no other nonzero singular value), $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1,0)^\top$, $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$. (b) $\mathrm{col}(A)$ is spanned by $\mathbf{u}_1$; $\mathrm{null}(A^\top)$ by $(1,-1,0)^\top$ and $\mathbf{e}_3$; $\mathrm{row}(A)$ by $\mathbf{v}_1$; and $\mathrm{null}(A)$ by $(1,-1)^\top$. (c) $A^+ = \frac{1}{2}\mathbf{v}_1\mathbf{u}_1^\top = \frac{1}{4}\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$.

11.3. (a) $AA^\top = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix}$ has eigenvalues 25 and 9, so $\sigma_1 = 5$, $\sigma_2 = 3$, $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$. (b) $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1,0)^\top$, $\mathbf{v}_2 = \frac{1}{3\sqrt{2}}(1,-1,4)^\top$, $\mathbf{v}_3 = \frac{1}{3}(2,-2,-1)^\top$. (c) $A_1 = \begin{pmatrix} 2.5 & 2.5 & 0 \\ 2.5 & 2.5 & 0 \end{pmatrix}$, $\|A - A_1\|_F = 3$, $\|A\|_2 = 5$, $\|A\|_F = \sqrt{34}$.

11.4. (a) $A^\top = V\Sigma^\top U^\top$. (b) $A^{-1} = V\Sigma^{-1}U^\top$. (c) $|\det(U)| = |\det(V)| = 1$. (d) $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$ has only one singular value; for the example, both norms are $\sqrt{5}$.

11.5. (a) $\lambda = \frac{2.01 \pm \sqrt{2.01^2 - 0.04}}{2}$, i.e., $\lambda_1 \approx 2.005$ and $\lambda_2 \approx 0.004988$. A is symmetric positive definite, so its singular values are its eigenvalues, and $\kappa(A) \approx 402$. (b) $\mathbf{x} = (1,1)^\top$ and $\mathbf{x} = (0,2)^\top$. The relative change of $\mathbf{b}$ is $\frac{0.01}{\|(2,2.01)\|} \approx 0.0035$, and the relative change of $\mathbf{x}$ is $\frac{\sqrt{2}}{\sqrt{2}} = 1$, an amplification by about $284 \leq \kappa(A)$. (c) $\kappa(A^\top A) = \kappa(A)^2 \approx 1.6 \cdot 10^5$: solving the normal equations of a least squares problem with this matrix would lose about 5 digits instead of 2.6.

11.6. (a) $S = \mathrm{diag}(2, 0.5)$; the principal components are $\mathbf{e}_1$ and $\mathbf{e}_2$; $\frac{2}{2.5} = 80\%$; the average reconstruction error is $0.5 = \lambda_2$. (b) The centered points are $(-2,-2)$, $(0,0)$, $(2,2)$, with eigenvalues $\frac{16}{3}$ and 0: the data lies exactly on a line, and the first principal component explains 100% of the variance. (c) The first right singular vector of the uncentered data points roughly towards the mean $(3,4)$, not in the direction in which the data varies.